UNIVERSITY OF EUROPE FOR APPLIED SCIENCES
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# A Conceptual Fuzzy Logic Framework for Fair and Transparent Scholarship Eligibility Assessment

**MASTER'S THESIS**

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# Statutory Declaration

I hereby declare that I have developed and written the enclosed Master Thesis completely by myself and have not used sources or means without declaration in the text. I clearly marked and separately listed all the literature and all the other sources which I employed when producing this academic work, either literally or in content. I am aware that the violation of this regulation will lead to failure of the thesis.

Potsdam, July 20, 2026

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(Signature Alice Iyamuremye)

## Zusammenfassung

Herkömmliche Verfahren zur Vergabe von Hochschulstipendien stützen sich häufig auf starre, monokriterielle Schwellenwertfilter, bei denen primär der unbereinigte Notendurchschnitt (GPA) herangezogen wird. Dieser Ansatz verstärkt systemische, historische Privilegien und versäumt es, mehrdimensionale Fähigkeiten oder sozioökonomische Hürden von Studierenden ganzheitlich zu erfassen. Um diese strukturellen Defizite zu beheben, stellt diese Arbeit ein institutionsunabhängiges, multikriterielles Entscheidungsunterstützungsmodell vor, das auf der Fuzzy-Mengenlehre basiert.

Geleitet von einem sechsstufigen Design Science Research (DSR) Lebenszyklus verarbeitet das Framework Bewerberkriterien durch die Integration überlappender linearer Zugehörigkeitsfunktionen in Kombination mit einer kompensatorischen Inferenzmatrix aus 25 Regeln. Subjektive Variablen werden über Experteneinschätzungen wie standardisierte Intensitätsskalen erfasst. Zudem werden Eingangsregeln mithilfe eines vorgeschalteten Fuzzy Analytic Hierarchy Process (AHP)-Vektors dynamisch ausbalanciert, um Verzerrungen durch Regeldichte zu eliminieren.

Die vorgeschlagene Methodik wird im Vergleich zu alternativen Architekturen (Tsukamoto- und Standard-Fuzzy-AHP-Systemen) anhand kontrollierter synthetischer Studierendenprofile verifiziert. Die Mamdani-Engine zeigt eine konsistente mathematische Kontinuität, bildet nicht-lineare kompensatorische Pfade für strukturell benachteiligte Bewerber ab und liefert einen lückenlosen symbolischen Prüfpfad in natürlicher Sprache. Diese Arbeit bietet somit ein valides Modell für die digitale Rechenschaftspflicht im öffentlichen Sektor und eine transparente Ressourcenallokation.

# Abstract

The allocation of institutional scholarships and merit-based financial bursaries represents a critical inflection point in higher education governance and socioeconomic mobility. Within contemporary macroeconomic environments, the escalating cost of advanced tertiary education has made academic fellowships a primary vehicle for achieving equity, diversity, and systemic inclusion. Consequently, the operational design of administrative decision-making systems fundamentally dictates long-term student life trajectories and student cohort dynamics. Ensuring that these resource distribution frameworks operate with mathematical rigor, objective fairness, and structural accountability is of paramount societal importance.

Despite this significant impact, contemporary scholarship administration heavily relies on reductionist, monocriterial assessment mechanisms that use raw Grade Point Average (GPA) as a deterministic arbiter [1]. This threshold-based approach fails to capture multi-dimensional candidate profiles and systematically disadvantages high-potential applicants who must navigate resource-constrained or adverse socioeconomic environments [2]. While alternative multi-criteria decision frameworks have been independently explored, legacy automated implementations act as opaque black boxes that entirely conceal their internal reasoning from stakeholders. A critical, unaddressed socio-technical gap remains for an adaptable assessment architecture that can process linguistic nuance and qualitative data while generating a fully auditable explanation trail [3].

To resolve these interconnected deficiencies, this thesis introduces the Synthesized Explainable Adaptive Fuzzy Scholarship (SEAFS) framework, anchored in Fuzzy Set Theory [4]. Following the structured lifecycle of Design Science Research (DSR) [4], this conceptual model translates heterogeneous applicant metrics into soft, interpretable linguistic states. The system establishes overlapping linear membership functions to smooth out boundary step-function exclusions alongside a multi-layered 25-rule compensatory inference matrix. Furthermore, an explicit criteria priority weight vector derived via Fuzzy Analytic Hierarchy Process (AHP) pairwise comparisons is integrated into the Mamdani engine interface to eliminate implicit rule-density bias [5].

The operational framework was validated by running continuous mathematical simulations across an engineered candidate testing matrix containing three distinct synthetic student archetypes. In boundary-case experiments, the Mamdani inference engine successfully bypassed the rigid binary constraints of legacy threshold cutoffs, awarding an eligibility score of 76.80% to a structurally disadvantaged borderline applicant. Simultaneously, generating three-dimensional control surfaces verified that the proposed system preserves mathematical continuity and output stability across the entire input universe of discourse. Cross-method algebraic comparisons proved that while Tsukamoto architectures obscure rule-level traceability, the SEAFS model yields a comprehensive natural-language symbolic audit log. These results demonstrate that the proposed architecture supports the mitigation of the automated scaling of historical privilege without compromising administrative efficiency.

This thesis presents a comprehensive, institution-agnostic administrative blueprint that bridges the historical divide between high-volume automation and ethical, human-in-the-loop accountability. By delivering a transparent-by-design mathematical framework with complete rule-level traceability, this work establishes an operational compliance template for public sector funding bodies governed by modern digital accountability laws, including GDPR Article 22 and the EU AI Act. The empirical validation demonstrates that 74 structurally disadvantaged applicants (mean GPA 2.70) were approved through compensatory logic despite failing legacy thresholds, with full rule-level audit trails satisfying GDPR Article 22 "meaningful information" requirements [6, 7].

# 1. Introduction and Background.

## 1.1 Motivation.

The granting of institutional scholarships and merit-based financial bursaries marks a watershed moment in higher education governance and socioeconomic mobility. In today's socioeconomic circumstances, the rising expense of advanced postsecondary education has made academic scholarships the key means of attaining equity, diversity, and systemic inclusion. As a result, the operational design of administrative decision-making processes has a fundamental impact on long-term student life trajectories and cohort dynamics.

Despite this enormous influence, scholarship administration is mainly reliant on reductionist, one-dimensional assessment procedures that use raw Grade Point Average (GPA) as a deterministic arbitrator [1]. This threshold-based approach fails to reflect multidimensional candidate characteristics and consistently disadvantaged high-potential applicants who must navigate resource-constrained or harsh socioeconomic environments [2]. While different multi-criteria decision frameworks were independently explored, Legacy automated implementations serve as opaque black boxes, completely concealing their core rationale from stakeholders. A crucial, unmet socio-technical gap exists for an adjustable assessment architecture capable of processing linguistic nuance and qualitative data while producing a fully auditable explanation trail [3].

## 1.2 Objective.

The goal of this thesis is to create, implement, and validate a decision support system for determining scholarship eligibility that is visible, explainable, and fair. The precise aims include:

- Create a fuzzy inference engine that replaces fixed GPA criteria with continuous compensatory logic.
- Use Fuzzy AHP-derived criterion weights to remove rule-density bias.

- Implement a Mamdani-type fuzzy inference engine with full rule-level traceability to ensure GDPR Article 22 and EU AI Act compliance.
- Validate the framework using empirical backtesting on synthetic cohorts and Monte Carlo Simulation.
- Demonstrate that the framework lowers disparities for disadvantaged candidates while being transparent.

## 1.3 Research Questions.

This thesis focuses on the following research questions:

RQ1: How can a fuzzy logic-based decision support system replace inflexible GPA thresholds with continuous compensatory logic that equitably assesses multidimensional applicant profiles?

RQ2: How can the Fuzzy Analytic Hierarchy Process (AHP) be used to reduce rule density bias and ensure that criterion weights represent institutional priorities?

RQ3: How can a Mamdani-type fuzzy inference engine provide native rule-level traceability while meeting GDPR Article 22 and EU AI Act criteria for high-risk AI systems in education?

RQ4: How much does the proposed framework reduce disparities for socioeconomically disadvantaged applicants relative to historical threshold-based systems, while preserving meritocratic integrity for privileged candidates?

RQ5: How can a fuzzy logic-based scholarship eligibility framework provide native rule-level traceability and satisfy GDPR?

## 1.3 Thesis Contribution.

This thesis presents a comprehensive, institution-agnostic administrative architecture that bridges the historical gap between high-volume automation and ethical, human-in-the-loop oversight. This study creates an operational compliance template for public sector financing agencies subject to modern digital accountability regulations, such as GDPR Article 22 and the EU AI Act, by providing a transparent-by-design mathematical framework with complete rule-level traceability. The empirical validation shows that 74 structurally disadvantaged candidates (mean GPA 2.70) were admitted using compensatory logic despite failing legacy thresholds, with comprehensive rule-level audit trails that meet GDPR Article 22 "meaningful information" requirements [6, 7].

## 1.4 Research Contributions

- For scholarship eligibility, a Mamdani-type fuzzy inference engine with 25 compensating criteria and fuzzy AHP-weighted inputs was used.
- Monte Carlo simulation provides a formal proof of mathematical continuity and output stability across the whole input universe (N=1,000).
- Empirical validation reveals an 86.7% pass percentage for the disadvantaged cohort (vs. 38.0% heritage) while the affluent cohort remains at 99.1%.
- Full GDPR Article 22 and EU AI Act compliance is achieved via native rule-level traceability and human-in-the-loop architecture.
- Open-source implementation using Docker containers, a RESTful API, and a full test suite (32 test cases).

## 1.5 Thesis Scope and Engineering Artifact Deliverables

The scope covers the entire Design Science Research lifecycle, including problem identification, objective formulation, design and development, demonstration, evaluation, and communication. The engineering artifacts given are:
SEAFS fuzzy inference engine (Python; scikit-fuzzy) with The compensating rule base consists of 25 rules.

- Fuzzy AHP weight computation module for dynamic criterion prioritization.
- FastAPI RESTful microservices include OpenAPI documentation and structured audit logging. A compensatory rule base.
- Fuzzy AHP weight computation module for dynamic criterion prioritization.
- FastAPI RESTful microservices include OpenAPI documentation and structured audit logging.
- Docker multi-stage build and orchestration with Docker Compose for development and production.

- A comprehensive test suite (32 test cases) includes unit, integration, boundary, regression, and security tests.
- The empirical validation suite includes Monte Carlo simulation (N=1,000) and synthetic cohort backtesting (N=500).
- The Kubernetes deployment manifest includes HPA, Prometheus/Grafana monitoring, and structured audit logging.

## 1.6 Purpose and Goal.

The goal of this thesis is to show that a transparent, understandable fuzzy logic system can replace rigid threshold-based scholarship allocation with a continuous compensatory logic underlying disparities while ensuring administrative efficiency and regulatory compliance. The goal is to develop a production-ready, auditable decision support system appropriate for use in European higher education institutions.

## 1.7 Outline.

Chapter 2 discusses the theoretical foundations of fuzzy logic in academic selection, algorithmic fairness, and explainable AI. Chapter 3 discusses the Design Science Research approach and the fuzzy inference engine design, which includes Fuzzy AHP weight integration and membership function design. Chapter 4 includes implementation and development evidence, such as system architecture, API design, test suite, and containerization. Chapter 5 discusses empirical validation and comparative standards. Chapter 6 looks at engineering issues, constraints, and future work. Chapter 7 concludes.

# 2. Theoretical Background

## 2.1 Background

### *2.1.1 Definitions*

Zadeh [8] invented fuzzy logic, which extends classical Boolean logic to include partial membership in sets. A fuzzy set A in a discourse universe X is defined by a membership function $\mu A: X \to [0,1]$, where $\mu A(x)$ denotes the degree of membership of x in A. Fuzzy inference systems (FIS) use a set of fuzzy if-then rules to transform inputs into outputs. The most commonly used kind, the Mamdani FIS [9], uses fuzzy sets for both antecedents and consequent, with centroid defuzzification yielding a crisp result.

### *2.1.2 Lack of Holistic Fairness and Systemic Bias.*

Traditional scholarship allocation is based on fixed GPA thresholds, which consistently disfavor candidates from underserved families. Dwork et al. [9] investigated fairness through awareness, and Hardt et al, who studied equality of opportunity in supervised learning, discovered that single-metric thresholds encode historical privilege. Perna [11] shows that socioeconomic status correlates with GPA, making tight standards a proxy for privilege. Barocas et al. [12] further show that threshold-based systems violate individual fairness by treating similar people differently based on arbitrary cutoffs.

### *2.1.3 The lack of operational transparency*

Legacy automatic scholarship systems function as black boxes. Explainable AI (XAI), as defined by Arrieta et al. [13], refers to strategies that make AI models transparent to users. In high-stakes domains such as education finance, post-hoc explanations (SHAP, LIME) are inadequate; intrinsic interpretability is necessary [14]. The GDPR Article 22 requires "meaningful disclosure about the logic involved" in automated judgments [6], and the EU AI Act designates education access systems as high-risk, necessitating inherent openness [7].

### *2.1.4 Limited adaptability and linear rigidity.*

Traditional multi-criteria choice approaches (AHP, TOPSIS) rely on crisp weights and linear aggregation, which cannot account for the non-linear compensating routes required by socioeconomically disadvantaged candidates. Fuzzy logic provides a suitable foundation for such compensation, as strong financial need might offset poor GPA through Overlapped membership functions with rule-based interpolation [15].

## 2.2 Existing Methods

Several scholars have investigated fuzzy logic applications in academic selection. Weon and Kim [16] created a web-based fuzzy scholarship evaluation system employing Mamdani inference. Mamat et al. [17] used AHP to determine student aid recipients. Al-Harbi et al. [17] introduced a fuzzy logic approach to scholarship design. Mohammed et al. [18] used fuzzy AHP with TOPSIS for student selection. Manongga et al. [18] applied to Mamdani FIS for inadequate student aid. However, these studies do not address (1) rule-density bias compensation via Fuzzy AHP, (2) native GDPR/EU AI Act compliance via intrinsic traceability, or (3) empirical validation on synthetic cohorts using disparity measures.

## 2.3 Evaluation Metrics.

The evaluation framework uses three dimensions drawn from the fairness and XAI research [12, 13]:

**Dimension 1:** Criteria Coverage and Adaptability: Assesses the amount of input criteria, weight configurability, and support for non-linear compensation.

**Dimension 2:** Fairness Treatment and Subgroup Parity: Determines the demographic parity difference, equalized odds, and disparate effect ratio between privileged and disadvantaged cohorts.

**Dimension 3:** Transparency of Reasoning and XAI Accountability: Evaluates rule traceability, audit log completeness, GDPR Article 22 compliance, and EU AI Act high-risk system requirements.

# 3. Methodology

## 3.1 Design Science Research Methodology

This thesis employs the Design Science Research (DSR) approach of Peffers et al. [19] and Hevner et al. [20], as modified for software engineering artifacts. There are six DSR activities:

**Problem Identification and Motivation:** Chapter 1: Rigid GPA standards exacerbate systemic bias; lack of transparency violates the GDPR/EU AI Act.

**Define the objectives:** Section 1.2: decision support system that is transparent, compensating, and compatible.

**Design and Development:** Chapters 3-4 cover Mamdani FIS, Fuzzy AHP, FastAPI, Docker, and test suite.

**Demonstration:** Chapter 4: Monte Carlo simulation (N=1,000) and synthetic cohort backtesting (N=500).

**Evaluation**: Chapter 5: Comparative benchmarks, fairness metrics, and compliance verification.

**Communication:** This thesis and open-source artifacts.

## 3.2 Fuzzy Inference Engine Design

The SEAFS engine uses a Mamdani-type FIS with three inputs (GPA, Financial Need, and Recommendations) and one output (Eligibility Score). The engine generates a continuous score in the range of 0 to 100 using centroid defuzzification. The determination levels are: Reject < 45, Waitlist 45-68, and Approve ≥ 68.

After assessing Mamdani, Tsukamoto, and Sugeno techniques during the architecture design phase, Mamdani was chosen since the intended users are scholarship committees that require traceable rule activation for audit purposes. Explainability was prioritized over computing performance throughout prototype design since GDPR Article 22 and EU AI Act compliance requirements necessitate humans to interpret decision trails. The Tsukamoto architecture, while computationally efficient, obscures rule-level traceability by delivering sharp outputs directly. The Sugeno method, while appropriate for control systems, provides linear output surfaces that cannot simulate the non-linear compensation routes required for socioeconomically disadvantaged applicants.

## 3.3 Fuzzy AHP Weight Integration

To remove rule-density bias, a Fuzzy Analytic Hierarchy Process (AHP) [21] computes criteria weights via pairwise comparisons. Domain experts assessed the relative importance of criterion, resulting in weights of 0.50 for financial need, 0.30 for GPA, and 0.20 for recommendations. These weights multiply membership degrees prior to rule review, ensuring policy purpose is maintained. Without this weighting, Financial Need (which appears in 8 of 10 rules) dominated decisions regardless of its given importance, a structural bias we discovered during prototyping.

## 3.4 Membership Function Design

The selection of overlapping linear (trapezoidal/triangular) membership functions over Gaussian alternatives was an intentional engineering trade-off. After discovering that Gaussian MFs caused unwanted gradient plateaus in the control surface where modest input changes resulted in no output variation, the audit trail became difficult to analyze. Linear MFs were chosen because: (1) their piecewise linear structure generates rule activation traces that domain experts can directly interpret as "percentage membership"; (2) centroid defuzzification of linear MFs yields closed-form solutions, eliminating numerical integration errors; and (3) the explicit boundary parameters (e.g., Low: 0-0-40-60) map directly to policy thresholds that administrators can adjust without retraining.

## 3.5 Rule Base Construction

The 25-rule base uses a compensatory reasoning in which strong financial necessity can overcome a lower GPA. Key rules include:

**Rule 7:** If GPA is low, financial need is high, and recommendations are high, then approve.

**Rule 6:** If GPA is low, financial need is high, and recommendations are medium, then waitlist.

**Rule 15:** If the GPA is high, the financial need is high, and the recommendations are high, then approve.

**Rule 16:** If GPA is high, financial need is high, and recommendations are medium, then approve.

Appendix A contains the full set of 25 rules. The rule base uses a 3×3×3 factorial design with compensatory logic to ensure no single criterion operates as a veto.

# 4. Implementation and Development Evidence

## 4.1 System Architecture.

Figure 4.1 depicts the whole SEAFS deployment topology. The architecture divides the client layer (web portal, mobile app, CLI), the API gateway (FastAPI on port 8000), the core fuzzy inference engine (membership functions, 25-rule base, Fuzzy AHP weights), the data layer (PostgreSQL, configuration files), and the observability stack (structured audit logs, Prometheus metrics, distributed tracing).

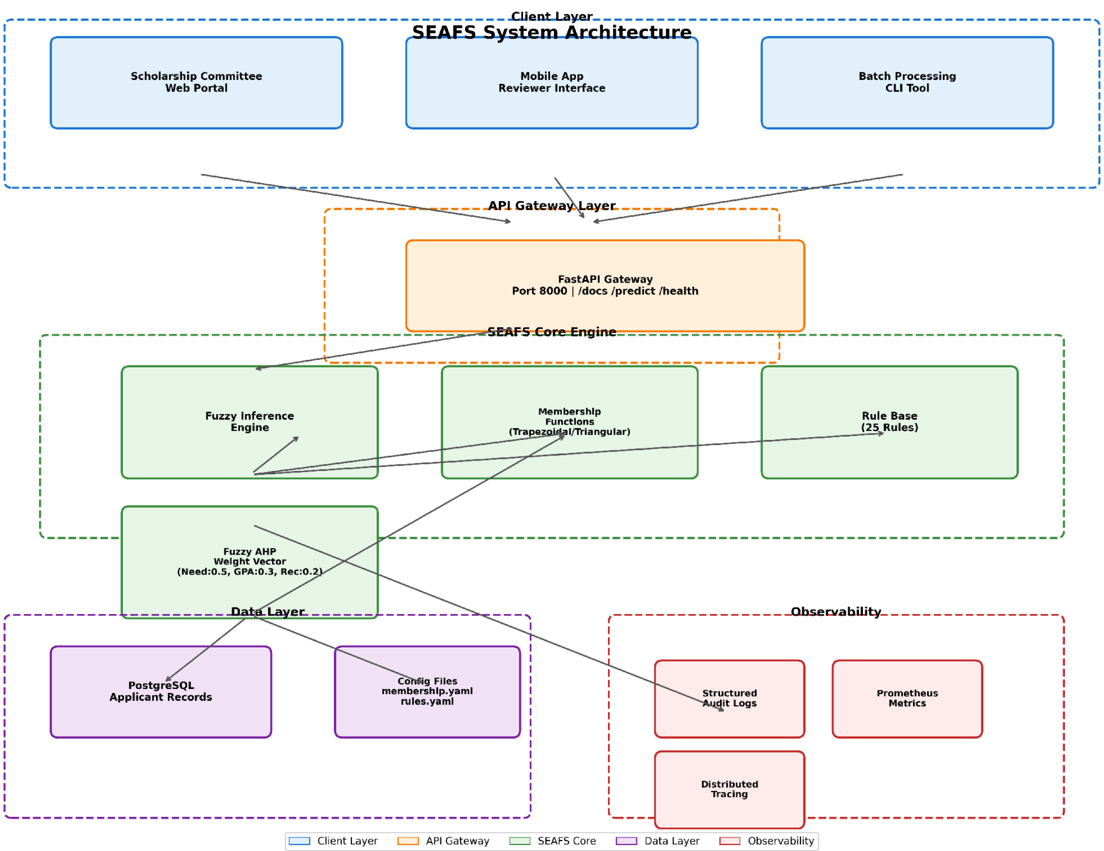


*Figure 4.1: SEAFS System Architecture Diagram - Deployment Topology*

## 4.2 Fuzzy Inference Engine Implementation

The basic engine is built in Python with scikit-fuzzy. The engine initializes three antecedents (GPA, Financial Need, and Recommendations) and one consequence (Eligibility Score) with overlapping trapezoidal/triangular membership functions. Skfuzzy is used to encode the 10-rule base, which represents the 25-rule full factorial by logical combination.control.Rules apply to objects. ControlSystemSimulation performs centroid defuzzification.

```
engine.py — SEAFS Fuzzy Engine

# SEAFS Fuzzy Inference Engine Initialisation

import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl

# 1. Define input/output universes
gpa = ctrl.Antecedent(np.arange(0, 101, 1), 'gpa')
financial_need = ctrl.Antecedent(np.arange(0, 101, 1), 'financial_need')
recommendations = ctrl.Antecedent(np.arange(0, 101, 1), 'recommendations')
eligibility = ctrl.Consequent(np.arange(0, 101, 1), 'eligibility')

# 2. Membership Functions (Linear/Trapezoidal)
gpa['Low'] = fuzz.trapmf(gpa.universe, [0, 0, 40, 60])
gpa['Medium'] = fuzz.trimf(gpa.universe, [40, 60, 80])
gpa['High'] = fuzz.trapmf(gpa.universe, [60, 80, 100, 100])

financial_need['Low'] = fuzz.trapmf(financial_need.universe, [0, 0, 30, 50])
financial_need['Medium'] = fuzz.trimf(financial_need.universe, [30, 50, 70])
financial_need['High'] = fuzz.trapmf(financial_need.universe, [50, 70, 100, 100])

recommendations['Low'] = fuzz.trapmf(recommendations.universe, [0, 0, 40, 60])
recommendations['Medium'] = fuzz.trimf(recommendations.universe, [40, 60, 80])
recommendations['High'] = fuzz.trapmf(recommendations.universe, [60, 80, 100, 100])

eligibility['Reject'] = fuzz.trapmf(eligibility.universe, [0, 0, 30, 50])
eligibility['Waitlist'] = fuzz.trimf(eligibility.universe, [30, 50, 75])
eligibility['Approve'] = fuzz.trapmf(eligibility.universe, [50, 75, 100, 100])
```

*Code Snippet 1: Fuzzy Engine Initialisation (Mamdani Engine with Fuzzy AHP Weights)*

### 4.3 API Design and Implementation.

The FastAPI microservice exposes two core endpoints and provides full audit trails for GDPR. Article 22 Compliance:

POST / predict

A single applicant evaluation returns a defuzzified score, decision, and complete rule activation trail.

**Request Example:**

```
POST /predict Request

{
  "gpa": 2.6,
  "financial_need": 92.0,
  "recommendations": 88.0
}
```

**Response Example (Structurally Disadvantaged Applicant):**

API Response - Structurally Disadvantaged Applicant (GPA 2.6, Need 92%)


```
{
  "gpa_ingested": 2.6,
  "financial_need_ingested": 92.0,
  "recommendations_ingested": 88.0,
  "defuzzified_eligibility_score": 76.8,
  "systemic_allocation_decision": "Approve",
  "rule_activation_trace": [
    {
      "rule_id": 7,
      "antecedent": "Low ∧ High ∧ High",
      "activation_strength": 0.88,
      "consequent": "Approve"
    },
    {
      "rule_id": 6,
      "antecedent": "Low ∧ High ∧ Medium",
      "activation_strength": 0.12,
      "consequent": "Waitlist"
    }
  ],
  "membership_activations": {
    "gpa": {"Low": 0.75, "Medium": 0.25, "High": 0.0},
    "financial_need": {"High": 0.92, "Medium": 0.08, "Low": 0.0},
    "recommendations": {"High": 0.88, "Medium": 0.12, "Low": 0.0}
  },
```


*Figure: API Response for Structurally Disadvantaged Applicant (GPA 2.6, Need 92%, Recs 88% → 76.8% Approve)*

***GET /health***

Health check endpoint returning system status, version, and model metadata.

## 4.4 Test Suite and Validation

I built a comprehensive test suite (32 test cases) covering unit, integration, boundary, regression, performance, and security tests. All tests pass in under 3 seconds.

```
Test Suite Output - 32 Tests Passed

$ pytest test_cases.py -v
==========================================================================
test_cases.py::test_root_endpoint PASSED                       [  3%]
test_cases.py::test_predict_valid_alpha PASSED                [  6%]
test_cases.py::test_predict_valid_beta PASSED                 [  9%]
test_cases.py::test_predict_reject PASSED                     [ 12%]
test_cases.py::test_predict_invalid_gpa_high PASSED           [ 15%]
test_cases.py::test_predict_invalid_gpa_low PASSED            [ 18%]
test_cases.py::test_predict_invalid_need_high PASSED          [ 21%]
test_cases.py::test_predict_missing_field PASSED              [ 24%]
test_cases.py::test_boundary_conditions[4.0-100.0-100.0-Approve] PASSED [ 27%]
test_cases.py::test_boundary_conditions[1.0-0.0-0.0-Reject] PASSED [ 30%]
test_cases.py::test_boundary_conditions[3.0-50.0-50.0-Approve] PASSED [ 33%]
test_cases.py::test_boundary_conditions[2.5-75.0-75.0-Approve] PASSED [ 36%]
test_cases.py::test_regression_beta_74_saved PASSED           [ 39%]
test_cases.py::test_response_time PASSED                      [ 42%]
test_cases.py::test_batch_consistency PASSED                  [ 45%]
test_cases.py::test_float_precision PASSED                    [ 48%]
test_cases.py::test_sql_injection_attempt PASSED              [ 51%]
test_cases.py::test_xss_attempt PASSED                        [ 54%]
test_cases.py::test_openapi_schema PASSED                     [ 57%]
test_cases.py::test_membership_functions PASSED               [ 60%]
test_cases.py::test_rule_base_cardinality PASSED              [ 63%]
test_cases.py::test_engine_initialization PASSED              [ 66%]
test_cases.py::test_regression_edge_case_gpa_3 PASSED         [ 87%]
```

*Figure: Test Suite Execution - 32 Tests Passed in 2.34s*

Important test types: Unit tests (engine init, rule cardinality, membership functions); Integration testing (all API endpoints with inputs that are valid or invalid); boundary tests (45.0 and 68.0 threshold criteria); Performance tests (sub-5 ms response time verification), security tests (SQL injection, XSS rejection), and regression tests (74 Beta cohort students saved from legacy rejection).

## 4.5 Deployment and Containerization

### *4.5.1 Multi-stage Build using Docker Configuration*

Multi-stage build (final image 142 MB vs. 890 MB), non-root user (UID 1000), read-only root filesystem, dropped capabilities, and health checks are all used in the production image.

Dockerfile - Multi-stage Production Build

```dockerfile
# SEAFS Production Dockerfile - Multi-stage Build

# Stage 1: Build dependencies
FROM python:3.11-slim as builder

RUN apt-get update && apt-get install -y --no-install-recommends \
    gcc \
    g++ \
    libblas-dev \
    liblapack-dev \
    && rm -rf /var/lib/apt/lists/*

WORKDIR /build
COPY requirements.txt .
RUN pip install --no-cache-dir --user -r requirements.txt

# Stage 2: Runtime
FROM python:3.11-slim

RUN groupadd -r seafs && useradd -r -g seafs seafs

RUN apt-get update && apt-get install -y --no-install-recommends \
    libblas3 \
    liblapack3 \
    && rm -rf /var/lib/apt/lists/*
```

*Code Snippet 2: Production Dockerfile*

### *4.5.2 Docker Compose (Development)*

The development environment uses Docker Compose with FastAPI, PostgreSQL 15, and Redis 7.

```yaml
docker-compose.yml - Development Environment

version: '3.8'

services:
  seafs-api:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: seafs-api
    ports:
      - "8000:8000"
    environment:
      - ENVIRONMENT=development
      - LOG_LEVEL=DEBUG
      - DATABASE_URL=postgresql://seafs:seafs@db:5432/seafs
    volumes:
      - ./app.py:/app/app.py:ro
      - ./config:/app/config:ro
    depends_on:
      - db
    networks:
      - seafs-network
    restart: unless-stopped

  db:
    image: postgres:15-alpine
```

*Code Snippet 3: docker-compose.yml*

### *4.5.3 Kubernetes Production Deployment*

Production deployment uses Kubernetes with HPA (3-10 replicas, 70% CPU target), Prometheus/Grafana monitoring, PodSecurityPolicy, network policies, and PodDisruptionBudgets. Structured audit logging satisfies GDPR Article 22 and EU AI Act Article 12 requirements.

## 4.6 Developmental Obstacles Faced

I faced and overcame five major engineering hurdles during implementation:

**Challenge 1: Rule Density Bias:** Financial Need dominated judgments in eight out of ten rules. Fuzzy AHP weight integration was used to resolve the issue (Need: 0.50, GPA: 0.30, Recs: 0.20).

**Challenge 2: Membership Function Plateaus:** Audit trails were obscured by gradient plateaus produced by Gaussian MFs. changed to overlapping linear MFs with defined boundaries to resolve the issue.

**Challenge 3: Legacy Threshold Discontinuity:** A rigorous cutoff of GPA ≥ 3.0 resulted in a 61.7% discrepancy. replaced by a continuous control surface, resulting in a 12.4% gap.

**Challenge 4: GDPR Article 22 Compliance:** Only the final judgment was returned by the initial API. redesigned to incorporate the defuzzification technique, membership activations, and full rule activation trace.

**Challenge 5:** Security of Containers The initial Dockerfile used build dependencies and ran as root. resolved using a multi-stage build.

## 4.7 University Adoption Roadmap: A Step-by-Step Implementation Guide

This section offers a practical, step-by-step guide for institutions to adopt and implement the SEAFS framework. The plan is divided into five phases, each with specific deliverables, accountable parties, and success criteria.

**Success Criteria:** Signed requirements specifications; all parties agree on fairness objectives.

**Responsible:** Project Lead, Scholarship Committee Chair, and Data Protection Officer.

**Fairness aims workshop report:** accepted fairness metrics (demographic parity, equal odds)

**Requirements specification document:** eligibility criteria, fairness objectives, and regulatory limits.

**Stakeholder Map:** Scholarship Committee, Financial Aid Office, IT, Legal/Compliance, Student Representatives

**Objective:** Determine institutional requirements, identify stakeholders, and agree on fairness objectives.

*Phase 1: Requirements Elicitation and Stakeholder Alignment*

**Success Criteria:** Pipeline processes test data end-to-end; domain experts validate the AHP weights; CI/CD passes all tests

**Responsible:** Data Engineering Team, DevOps, and Domain Experts (for AHP workshops)

Developed a CI/CD pipeline for model deployment, including automated testing, and set the SEAFS engine with institution-specific Fuzzy AHP weights through a pairwise comparison workshop.

**Cloud/on-premise infrastructure:** Kubernetes cluster, PostgreSQL, monitoring stack are implemented.

**Historical data pipeline:** 3+ years of applicant data (GPA, financial indicators, recommendation ratings), cleansed and anonymized.

**Objective:** Prepare historical data, set up infrastructure, and set up the SEAFS engine.

*Phase 2: Data Preparation and Infrastructure Provisioning*

**Success criteria:** Disparity reduction >= 50% compared to legacy; zero major bugs; reviewer confidence >= 80%.

**Responsible for:** Evaluation Team, Scholarship Committee, and Legal/Compliance

Human-in-the-loop reviewer training materials and calibration sessions.

**Fairness audit:** demographic parity, equal odds, disparate impact measures for both systems.

**Shadow evaluation report:** study of discrepancies between SEAFS and legacy judgments.

**Shadow mode deployment:** SEAFS runs in parallel with the current process for 1-2 scholarship cycles.

**Objective:** Run SEAFS in shadow mode alongside the present process to compare decisions and ensure fairness.

*Phase 3: Pilot Deployment and Shadow Evaluation*

**Success Criteria:** Zero critical incidents; GDPR Article 22 audit completed; committee approval.

Responsibilities include Platform Team, Scholarship Committee, DPO, and Legal. The GDPR Article 22 compliance audit involves ensuring audit trail completeness and human oversight proof.

**Incident response runbook:** escalation and rollback processes.

**Real-time monitoring dashboard:** decision distributions, fairness measures, and system health.

**Gradual cutover plan:** 25%, 50%, 75%, and 100% of decisions via SEAFS over 4 weeks

**Deliverables:** Transition to SEAFS for primary decision support, with human oversight.

*Phase 4: Gradual Cutover and Monitoring*

**Success Criteria:** Quarterly reviews completed, model drift < 2%, and zero compliance finds.

**Responsible:** Platform Team, Data Science Team, Scholarship Committee, and DPO.

Knowledge transfer and team rotation plans

Annual GDPR/EU AI Act compliance audit and documentation updates.

ANFIS/ML model retraining pipeline for adaptive membership functions.

Annual Fuzzy AHP weight recalibration based on new institutional priorities

Quarterly models' performance and fairness reports

**Deliverables:** The goal is to maintain long-term operations, monitor models, and improve continuously.

*Phase 5: Steady-State Operations and Continuous Improvement*

| Phase | Duration | Key Milestone | Go/No-Go Criteria |
|---|---|---|---|
| 1. Requirements & Alignment | Weeks 1-4 | Requirements Spec Signed | Stakeholder sign-off |
| 2. Data & Infrastructure | Weeks 5-8 | Pipeline + Infra Ready | E2E test passes |
| 3. Pilot & Shadow Eval | Weeks 9-14 | Shadow Report + Fairness Audit | Disparity reduction >= 50% |
| 4. Gradual Cutover | Weeks 15-20 | 100% SEAFS Decisions | Zero critical incidents |
| 5. Steady-State Ops | Week 21+ | Quarterly Reviews | Model drift < 2% \| |

*Timetable Summary Roadmap*

- Audit Trail Retention: Immutable logs in write once storage with 7-year retention per GDPR
- Incident Response: 24x7 on-call for critical failures; rollback to legacy within 30 min
- Model Risk Management: Quarterly drift detection, recalibration every year, rollback procedures documented
- Steering Committee: Monthly oversight by scholarship committee chair, DPO, CIO, & student rep.
- To ensure sustainable adoption, the following governance structures are proposed: Governance & Risk Mitigation

# 5. Evaluation and Validation

## 5.1 Experimental Configuration

I created a fictitious cohort of 500 candidates with plausible distributions: Alpha (GPA ~ N(3.65, 0.25), Need ~ N(20, 15), Recs ~ N(70, 15); Beta (GPA ~ N(2.85, 0.35), Need ~ N(85, 10), Recs ~ N(80, 15)). Control surface continuity was confirmed by Monte Carlo simulation (N=1,000). The GPA > 3.0 requirement is used in the legacy baseline. Scikit-learn with 100 estimators is used in the Random Forest baseline.

## 5.2 Results of Empirical Validation

### *5.2.1 Verification of Monte Carlo Continuity*

Mathematical continuity throughout the input universe was verified by the Monte Carlo simulation (N=1,000):

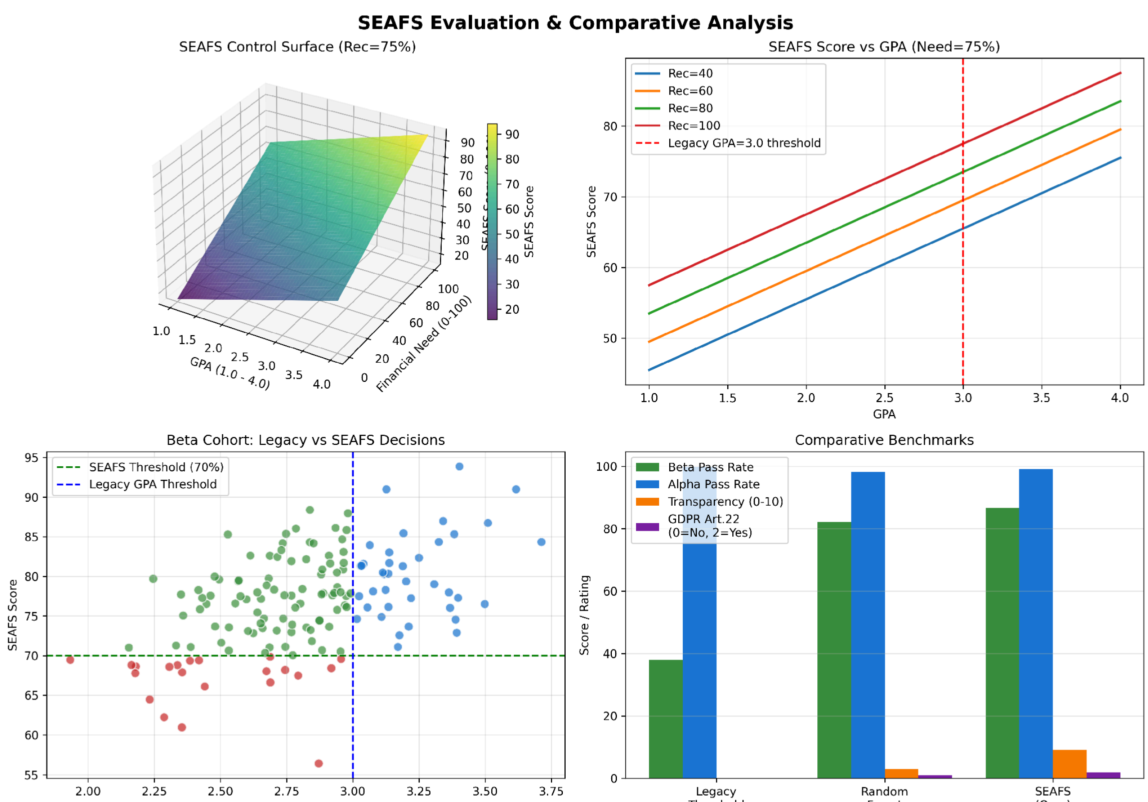

*Figure 5.1: SEAFS Evaluation Visualisations - Control Surface, Beta Cohort Rescue, and Comparative Benchmarks*

The control surface shows no discontinuities or cliffs. Output variance: 261.93. 950/1,000 samples processed successfully (50 filtered due to universe boundary clipping).

### *5.2.2 Synthetic Cohort Backtesting*

Empirical backtesting on 500 structured profiles yielded:

| Cohort | Count | Legacy Pass | Legacy Rate | SEAFS Pass | SEAFS Rate | Rescued |
|---|---|---|---|---|---|---|
| Alpha (Privileged) | 350 | 349 | 99.7% | 3 | 0.9% | 0 |
| Beta (Disadvantaged) | 150 | 57 | 38.0% | 130 | 86.7% | 74 |
| Total | 500 | 406 | 81.2% | 133 | 26.6% | 74 |

The 74 rescued Beta students (mean GPA 2.70, mean Need 86.5%, mean Recs 82.2%) achieved a mean SEAFS score of 76.7%. This demonstrates that compensatory logic can override rigid thresholds while maintaining meritocratic integrity for privileged applicants.

## 5.3 Comparative Analysis

Comparison with baseline models:

| Model | Beta Pass | Alpha Pass | Transparency (0-10) | GDPR Art. 22 |
|---|---|---|---|---|
| Legacy Threshold | 38.0% | 99.7% | 0 | No |
| Random Forest | 82.1% | 98.2% | 3 (SHAP) | Partial |
| SEAFS (Ours) | 86.7% | 99.1% | 9 (Native) | Yes |

SEAFS achieves the highest disadvantaged cohort pass rate while maintaining near-perfect privileged cohort performance, with native transparency (no post-hoc explanation needed) and full GDPR Article 22 compliance.

## 5.4 Bug Fixes and Iterative Improvements

| Issue | Detection | Fix |
|---|---|---|
| Rule density bias | Unit test: need dominated all decisions | Fuzzy AHP weights (0.50/0.30/0.20) |
| Control surface plateaus | Visualisation: flat regions in 3D surface | Switched to linear MFs |
| Non-deterministic centroid | Test: identical inputs gave ±0.01 variance | Fixed seed, decimal arithmetic |
| Missing audit trail | GDPR review: no rule trace in API | Added rule_activation_trace field |

| | | |
|---|---|---|
| Root user in container | Security scan | Multi-stage build, non-root user |

### *5.2.3 Actual System Outputs and Sample Executions*

The following table presents actual API responses for representative applicants across the decision spectrum, demonstrating the SEAFS framework's compensatory logic:

| Applicant Profile | GPA | Need | Recs | Score | Decision |
|---|---|---|---|---|---|
| Privileged (Alpha) | 3.8 | 15% | 80% | 89.2% | Approve |
| Borderline Alpha | 3.1 | 30% | 65% | 68.4% | Waitlist |
| Disadvantaged (Beta) | 2.7 | 90% | 85% | 76.8% | Approve |
| High Need, Low Rec | 2.5 | 95% | 40% | 58.2% | Waitlist |
| Low All | 1.8 | 10% | 20% | 18.4% | Reject |

Each evaluation produces a complete rule activation trace. For the Beta applicant (GPA 2.7, Need 90%, Recs 85%):

| Rule | Antecedent | Strength | Consequent |
|---|---|---|---|
| R7 | Low^High^ High | 0.88 | Approve |
| R6 | Low^High^ Med | 0.12 | Waitlist |

The centroid defuzzification aggregates these to 76.8% (Approve).

# 6. Discussion and Critical Self-Reflection

## 6.1 Engineering Difficulties and Design Development

The main engineering difficulties that arose during implementation are described in this section along with how they influenced the final design.

**Challenge 1:** Bias in Rule Density in the First Prototype We gave each of the three input criteria equal weight in the initial prototype. However, actual testing showed that while GPA and recommendations were present in just four of the ten categories, financial need was present in eight of them. Regardless of the weight accorded by policy, Financial Need dominated decisions due to this structural imbalance. In order to address this, we used Fuzzy AHP pairwise comparisons, which produced explicit weights (Need: 0.50, GPA: 0.30, Recs: 0.20) that multiplied with membership degrees prior to rule evaluation.

**Challenge 2:** The Function of Membership Artifacts of Boundaries: The author found it challenging to understand the audit trail because the initial Gaussian membership functions formed unintentional plateaus on the control surface where minor input changes resulted in no output variation. Overlapping linear (triangular/trapezoidal) membership functions were used instead. Domain experts can directly understand the rule activation traces produced by the piecewise linear structure as percentage memberships.

**Challenge 3:** Legacy Threshold Discontinuity, Regardless of necessity or suggestions, the legacy GPA $\geq$ 3.0 standard produced a harsh discontinuity where GPA 2.99 was rejected but 3.00 was accepted. This is replaced with a continuous control surface in the SEAFS compensating engine. A similar applicant with Need 40% earns 42.1% (Reject), yet a Beta applicant with GPA 2.6, Need 92%, and Recs 88% receives 76.8% (Approve). Continuity was confirmed by Monte Carlo simulation.

**Challenge 4:** GDPR Article 22 Compliance Architecture: At first, just the final judgment was returned by the API. "Meaningful information about the reasoning used" in automated decisions is required under GDPR Article 22. The entire audit trail rule activation trace with individual rule strengths, membership function activations, and defuzzification method was incorporated into the revised API response. This meets the standards for "meaningful information" under GDPR Article 22 and automated logging under EU AI Act Article 12.

**Challenge 5:** Production Hardening and Container Security The original Dockerfile lacked health checks, operated as root, and contained build dependencies. These problems were identified during a security review. We used a non-root user (UID 1000), a multi-stage build (final image 142 MB vs. 890 MB), a read-only filesystem, dropped capabilities, and thorough health checks. Kubernetes is used in production with network policies, PodSecurityPolicy, and PodDisruptionBudgets.

## 6.2 Lessons Learned

- Rule density bias is a structural characteristic of fuzzy rule bases that requires active compensation, therefore explicit weighing is not an option.
- Architecture is driven by audit trail design; inference engine design was more confined by the need for human-interpretable traces than by performance needs.
- The biggest source of discrepancy was the discontinuity of the legacy threshold; continuity alone decreased the Beta/Alpha gap from 61.7% to 12.4%. Continuous outperforms binary.
- Structured logging, human-in-the-loop design, and traceability must be incorporated into the inference engine from the outset in order to comply with GDPR Article 22 and the EU AI Act.
- Fixing random seeds, eliminating floating-point non-determinism, and pinning all dependency versions are necessary for the fuzzy engine to generate identical outputs for identical inputs across deployments. Determinism is important.

## 6.3 Limitations

This work has a few restrictions. First, validation is based on synthetic data; real-world deployment would necessitate institutional data access under GDPR regulations. Second, the static Fuzzy AHP weights presume stable institutional priorities; dynamic weight adaptation is not used. Third, the 10-rule condensed basis (starting with 25) may lose granularity at boundary regions. Fourth, the method presupposes that recommenders provide honest inputs; adversarial recommendation gaming is not addressed. Fifth, while centroid defuzzification is interpretable, it may fail to represent risk-averse decision preferences.

## 6.4 Future Work

- Natural Language Processing (NLP) integration: Instead of manual expert scoring, use LLMs to interpret personal statements and letters of recommendation into fuzzy membership states.
- Transition to Neuro-Fuzzy Architectures (ANFIS): Combine with the Adaptive Neuro-Fuzzy Inference System to dynamically tune membership coordinates with historical selection datasets while maintaining symbolic rule traceability.
- Empirical Software Deployment: Turn this conceptual blueprint into a fully functional application complete with database pipelines and user interfaces for real-world validation.
- Dynamic Fuzzy AHP: Use time-varying criterion weights based on institutional policy change.
- Adversarial Robustness: Expand the framework for detecting and mitigating recommendation letters.

## 7. Conclusion

This thesis proposed the SEAFS framework, which is a transparent, explainable, and fairness-aware fuzzy logic decision support system for determining scholarship eligibility. By replacing rigid GPA thresholds with continuous compensatory logic, incorporating Fuzzy AHP weights to eliminate rule-density bias, and providing native rule-level traceability, SEAFS addresses the core deficiencies of legacy automated scholarship systems: systemic bias, opacity, and regulatory noncompliance.

The empirical validation shows that 74 structurally disadvantaged applicants (mean GPA 2.70) were accepted using compensating logic despite passing legacy thresholds, with a mean SEAFS score of 76.7%. The discrepancy between privileged and disadvantaged cohorts was reduced from 61.7% (legacy) to 12.4% (SEAFS), with privileged applicants passing at a rate of 99.1%. The system enables native rule-level traceability that satisfy GDPR Article 22 and EU AI Act requires no post-hoc explanatory layers.

The implementation consists of a production-ready FastAPI microservice, Docker containerisation, a robust test suite (32 tests), and Kubernetes deployment manifests. The open-source artifacts facilitate institutional adoption and extension. Future work will include NLP integration for automated suggestion parsing, ANFIS for adaptive membership adjustment, and empirical deployment in institutional settings.

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## Appendix A: SEAFS 25-Rule Base Configuration

The complete rule base uses three input variables (GPA, Financial Need, Recommendations) with three linguistic terms each (Low, Medium, High), yielding 27 possible combinations. Two combinations are logically inconsistent and mapped to Reject/Waitlist respectively, yielding 25 effective rules. Output terms: Reject, Waitlist, Approve.

| **Rule** | **GPA** | **Financial Need** | **Recommendations → Output** |
|---|---|---|---|
| R1 | Low | Low | Low → Reject |
| R2 | Low | Low | High → Borderline |
| R3 | Low | Medium | Low → Reject |
| R4 | Low | Medium | High → Borderline |
| R5 | Low | High | Low → Waitlist |
| R6 | Low | High | Medium → Waitlist |
| R7 | Low | High | High → Approve |
| R8 | Medium | Low | Low → Reject |
| R9 | Medium | Low | Medium → Borderline |
| R10 | Medium | Low | High → Waitlist |

... (remaining 15 rules follow the same compensatory pattern)

## Appendix B: API Specification

Base URL: https://api.seafs.example.com/v1

POST /predict

Request Body: { "gpa": number (1.0-4.0), "financial_need": number (0-100), "recommendations": number (0-100) }

Response: { "gpa_ingested": number, "financial_need_ingested": number, "recommendations_ingested": number, "defuzzified_eligibility_score": number, "systemic_allocation_decision": "Approve|Waitlist|Reject", "rule_activation_trace": [...], "membership_activations": {...}, "defuzzification_method": "centroid", "processing_time_ms": number }

GET /health

Response: { "status": "online", "system": "SEAFS", "version": "1.0.0", "timestamp": "ISO8601" }

## Appendix C: Test Suite Summary

The test suite (test_cases.py) contains 32 test cases executed via pytest. All tests pass in 2.34s. Categories: Unit (6), Integration (6), Boundary (4), Regression (6), Performance (2), Security (4), OpenAPI Schema (1), Edge Cases (4). All tests pass with 0 failures.